MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPES 6AT6 AND 12AT6

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Triode, miniature, duplex diode, high-Mu

Outline	---	5-2 (EIA)
Base	---	E7-1
Envelope	---	T5-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	g	k	h	h	2a (diode)	1a (diode)	a (triode)

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	E1b	E2b	I1b	I2b	(B) Ehk	(B) Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	mAdc	v	ft
Maximum:									
Type 6AT6	6.9	330	---	---	---	1.0	1.0	100	(See
Type 12AT6	13.9	330	---	---	---	1.0	1.0	100	note 1)
Minimum:									
Type 6AT6	5.7	---	---	---	---	---	---	---	---
Type 12AT6	11.3	---	---	---	---	---	---	---	---
TEST CONDITIONS:									
Type 6AT6	6.3	250	-3	0	0	---	---	---	---
Type 12AT6	12.6	250	-3	0	0	---	---	---	---

GENERAL:

Qualification - Not required

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	See note 2	0.65	II	Ic	---	-0.5	µAdc
(B)1321	AC amplification	Ebb = 100 Vdc; Eg = 0.2 Vrms; Ecc = 0	0.65	II	Ep	7.7	---	Vac
1246	Audio frequency noise	Esig = 70 mVac; Rp = 0.2 Meg	0.65	II	EB	---	17	vu
1231	Emission (triode)	Eb = Ec = 30 Vdc (see note 2)	0.65	II	Is	25	---	mAdc
(B)1231	Emission (diode)	Eb = 10 Vdc	0.65 0.65	II II	I1s I2s	0.8 0.8	--- ---	mAdc mAdc
(B)1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Rp = 2,000 ohms	---	---	Ep	---	150	mVac
1301	Heater current							
	Type 6AT6		---	---	If	275	325	mA
	Type 12AT6		---	---	If	138	162	mA
1336	Heater-cathode leakage		---	---	Ihk	---	20	µAdc
1256	Electrode current (anode) (triode)		---	---	Ib	0.5	1.8	mAdc
1306	Transconductance		---	---	Sm	900	1,700	µmhos
(B)1316	Amplification factor		---	---	Mu	57	83	---
(B)1211	Insulation of electrodes		4.0	S3	---	---	---	---
(B)1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	Group A; Ehk = 100 Vdc; I1b = I2b = 1.0 mAdc; Rg = 0.05 Meg (min)	---	---	---	---	---	---
---	Life-test end points (500 hours)	AC amplification	---	---	Ep	6.0	---	Vac
		and Emission (diode)	---	---	I1s	0.3	---	mAdc
			---	---	I2s	0.3	---	mAdc

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
2. This test to be performed at the conclusion of the holding period.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL, MI
Navy -
Air Force - 11, 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

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